

1. Purpose

This document captures the evidence and decisions that define RecoverAI before implementation. It is the source of truth for why the project exists, what problem it solves, what the Buildathon requires, and what must not be built.

2. Project decision

Build RecoverAI for the AI Revenue Recovery track.

Exact product:

RecoverAI — Payment Failure Recovery Control Loop

One-line thesis:

RecoverAI turns a failed Razorpay payment into a bounded recovery decision: diagnose the failure, estimate recoverability, select the safest high-value intervention, execute only within merchant policy, verify the outcome through payment events, and stop or escalate with a complete audit trail.

3. Why this project

The project is strongest when evaluated as an engineering audition rather than an ideation exercise. The useful signal is the ability to turn a messy financial workflow into a reliable system with:

- narrow scope;
- real payment events;
- correct API behavior;
- probabilistic AI used where useful;
- deterministic financial controls;
- safe tool execution;
- explicit failure handling;
- measurable improvement;
- reproducible evaluation.

The uploaded research identifies RecoverAI as the strongest option because it combines Razorpay-native workflows, measurable economic impact, agentic execution, backend/reliability depth, and a credible benchmark.

4. Buildathon alignment

The research identifies the Revenue Recovery track as the best fit. The target behavior is not merely identifying failed payments or generating messages. The system should detect a recovery opportunity, choose and execute a bounded action, measure recovered money, apply stopping rules, and preserve an audit trail.

Important distinction:

- **Fact:** the Buildathon requires a working submission and the Revenue Recovery track emphasizes recovery, bounded execution, measurement, escalation/stopping rules, and auditability.
- **Inference:** hiring signal is strongest when the project demonstrates engineering judgment, reliability, safe automation, honest measurement, and ownership of the complete loop.

Do not treat inferred judging preferences as official scoring weights.

5. Razorpay surface to use

Primary integration surface:

1. Payments / payment failure context.
2. Orders to associate payment attempts with a commercial transaction.
3. Payment Links for a bounded recovery action.
4. Webhooks for asynchronous event-driven verification.
5. Test Mode for all money-related demonstrations.

Optional only after the core loop works:

- Subscriptions.
- Invoices.

Explicitly avoid expanding into refunds, disputes, RazorpayX, voice, WhatsApp, CRM integrations, or multi-channel orchestration for the MVP.

6. Reliability implications

The research highlights several integration requirements that must shape the design from day one:

- Webhooks are at-least-once delivery, so duplicate events must be safely ignored.
- Webhook signatures must be verified.
- Webhook handlers should acknowledge quickly and perform work asynchronously where appropriate.
- Tool calls must be bounded and idempotent where possible.

- The LLM must never receive direct unrestricted access to payment APIs.
- Payment state must be authoritative from verified payment events, not from model output.

7. Core product insight

The valuable problem is not “write a better payment reminder.” It is:

Given a failed payment and its context, what should happen next, if anything, and how do we know whether that decision was correct?

The system therefore needs five distinct responsibilities:

1. **Diagnosis:** What likely caused the failure?
2. **Prioritization:** Is recovery worth pursuing now?
3. **Decision:** Which approved action has the best expected value and lowest friction?
4. **Control:** Is the action permitted by deterministic merchant policy?
5. **Verification:** Did the payment actually recover?

8. AI boundary

AI is appropriate for:

- interpreting messy failure descriptions;
- synthesizing payment, order, customer, timing, and historical context;
- producing structured diagnosis;
- estimating qualitative/action-specific recovery likelihood;
- selecting among already-approved action classes;
- explaining the recommendation.

AI is not authoritative for:

- monetary limits;
- consent state;
- opt-out state;
- retry limits;
- whether an action is allowed;
- refund decisions;
- irreversible actions;
- final payment state.

The architecture must enforce this boundary in code.

9. Agent definition

RecoverAI is agentic because it performs a stateful closed loop:

event → context → diagnosis → decision → policy check → tool execution
→ event verification → next state

It is not a multi-agent system. One decision agent plus deterministic services is sufficient.

10. Benchmark thesis

The benchmark must answer:

Does the hybrid RecoverAI policy recover more simulated revenue, with fewer unnecessary interventions and no critical policy violations, than a reasonable fixed-rule baseline?

Compare at minimum:

1. Fixed-rule baseline.
2. LLM-only policy.
3. RecoverAI hybrid policy.
4. Policy-only ablation.
5. Optional LLM-without-guardrails safety ablation.

The simulator must generate hidden ground truth from latent customer/payment states. The observable inputs given to the system must not directly expose the answer.

11. Primary metrics

- Total simulated recovered amount.
- Incremental recovery versus fixed-rule baseline.
- Recovery rate.
- Expected recovered amount.
- Cost-adjusted recovered amount.

Safety metrics:

- Policy violation rate.
- Action-after-success rate.
- Hard-decline retry rate.
- Unnecessary intervention rate.
- Repeated-contact/intervention rate.
- Escalation rate.

Operational metrics:

- Cases processed.
- Median time to action.
- Average interventions per case.
- Tool-call failure rate.
- Duplicate-event suppression rate.
- Structured-decision validity rate.

12. Initial success thresholds

Use these as engineering targets, not claims about production performance:

- $\geq 10\%$ higher simulated recovered amount than fixed-rule baseline.
- 0 critical policy violations in the final benchmark.
- $< 5\%$ action-after-success errors.
- $< 10\%$ unnecessary interventions.
- 100% duplicate-event suppression in dedicated tests.
- $\geq 95\%$ valid structured decisions.

If the hybrid system does not outperform the rule baseline, report that result honestly. A valid conclusion may be that AI improves contextual diagnosis/explanation but deterministic rules are better for certain known failure classes.

13. Main risks

Risk 1: Duplicate of existing Razorpay recovery capabilities

Countermeasure: Compete on transparent decision quality, bounded control, benchmark evidence, and auditability, not on claiming a novel reminder mechanism.

Risk 2: Synthetic benchmark credibility

Countermeasure: Hidden latent state, seeded reproducibility, held-out evaluation, multiple baselines, explicit limitations, and no claim that simulation predicts production recovery.

Risk 3: LLM adds no real value

Countermeasure: Treat ablation as a first-class result. Do not force AI into deterministic decisions where it is unnecessary.

Risk 4: Integration uncertainty

Countermeasure: Build a provider adapter and simulator so the product remains demonstrable even if Test Mode coverage is limited.

Risk 5: Scope explosion

Countermeasure: Payment failures + Payment Links only for the core. Everything else is cut until the full loop is working.

14. Non-goals

Do not build:

- generic collections;
- B2B receivables;
- abandoned-cart recovery;
- standalone subscription recovery;
- voice or WhatsApp workflows;
- multi-agent orchestration;
- dynamic discounts;
- autonomous payment-method routing;
- live customer communication infrastructure;
- production money movement;
- unrestricted LLM tool use.

15. Decision rule

When choosing between a new feature and more reliability:

Choose reliability.

When behind schedule, cut breadth before cutting:

- benchmark;
- deterministic policy;
- webhook verification;
- audit trail;
- end-to-end recovery demonstration.

16. Evidence status

This document is grounded in the supplied Buildathon research and the current Linear project definition. Claims explicitly identified as inferences must remain labeled as such. Unknown Buildathon rules must not be invented.

